

AWS Athena

04-data-engineering/athena

Overview

Athena is a serverless interactive query service for data stored in S3. It lets you run SQL over files without provisioning or managing a dedicated analytics cluster.

It is valuable in ML workflows because teams need a fast way to inspect datasets, validate assumptions, sample training windows, and investigate predictions without building separate infrastructure first.

Core Concepts

- Athena uses schema on read, usually through the Glue Data Catalog, which means the meaning of the data is defined at query time rather than baked into a database load step.
- Performance and cost depend heavily on file format, compression, partitioning, and how much data each query scans.
- Workgroups, query result locations, and access controls help separate exploratory use from governed production style use.

How It Works

- Data is stored in S3, a table definition describes where it lives and how it is structured, and Athena executes SQL against that definition using an underlying distributed engine.
- Query results are written back to S3 and can be inspected directly, consumed by notebooks, or used to feed later data processing or reporting steps.
- Athena is often used iteratively: explore raw data, refine filters, validate counts, and then codify the proven logic in a more formal pipeline if it becomes recurring.

AI and ML Relevance

- Athena is useful for exploring labeling quality, checking class imbalance, auditing inference logs, and quickly slicing large event datasets before training decisions are made.
- It also complements Glue well, because cataloged curated datasets become directly queryable for both analysts and ML engineers.

Operational Notes

- Small files and poor partitions make Athena expensive and slow. File layout is not an implementation detail; it is part of performance design.
- Prefer columnar formats like Parquet for analytical workloads so only relevant columns are scanned.
- Athena is great for exploration and light operational analytics, but not a replacement for every transactional or low latency query requirement.

What To Remember

- Serverless SQL is convenient, but the data still needs structure and discipline.
- You pay largely for scanned data, so data layout drives cost.
- Athena is often the shortest path from raw S3 data to a useful answer during ML investigation.